

Nice — below is a complete, **paper-by-paper reading & implementation sequence** plus an exhaustive, step-by-step project plan for your *Volatility Forecasting* project (comparative analysis of traditional vs. modern hybrid ML+HAR approaches). I order the papers so you **read theory → simple baselines → evaluation → advanced ML / hybrid models → multivariate/graph/GP → experiments & write-up**. I include the **key equations / concepts to extract from each paper, concrete replication notes, which code repos exist to steal/extend**, and an actionable experiment matrix and repo structure you can copy into GitHub.

All referenced papers and code are linked in-line (click the citations). Read them in the order below, implement as you go, and use the experiment matrix (near the end) as your lab notebook.

Overview — high level plan

1. **Foundations / data** — learn realized measures, how to compute them and why they matter. (Andersen et al., Corsi) ([Economics Department](#))
 2. **Simple baselines** — HAR, GARCH, Realized-GARCH; compute realized variance (RV) and simple HAR regressions. ([statmath.wu.ac.at](#))
 3. **Evaluation machinery** — robust loss functions (QLIKE), Diebold-Mariano tests, MCS, VaR backtests. Use QLIKE/MSE and economic measures (VaR hedging P&L).
 4. **Modern / hybrid models** — HARNet (dilated CNN HAR hybrid), DeepVol (dilated causal convs on HF), Volt (GP stochastic-vol framework), GNNHAR (multivariate graph + HAR), Multi-Transformer and other Transformer variants. Replicate & compare. ([arXiv](#))
 5. **Rigorous experiments** — out-of-sample rolling windows, statistical tests, economic evaluation, ablation studies (HF vs RV; QLIKE vs MSE; pretrain vs from scratch).
 6. **Original contribution** — hybridize HARNet/DeepVol with Volt's uncertainty; add spillover GNN features; pretrain a universal volatility model; or propose a calibrated ensemble that improves risk-adjusted hedging. (Concrete extension ideas are interleaved below.)
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Reading & implementation sequence (exact order, with what to extract & code to run)

Read each paper once for ideas/intuition, then a second time to extract equations/algorithms and planning of replication. Implement minimal reproductions as you read.

I mark for each item: **What to learn**, **Key equations / highlights**, and **Code & data to use**.

1) Andersen, Bollerslev, Diebold & Labys — *Modeling and Forecasting Realized Volatility* (foundations on realized measures) ([Economics Department](#))

What to learn (first read):

- Why realized measures (RV, realized kernels) are *model-free* proxies of latent volatility and how high-frequency aggregation reduces measurement error.
- Practicalities: sampling frequency tradeoffs (microstructure noise), pre-processing tick/trade data to 1–5 minute returns, and robust realized estimators.

Key equations / items:

- Realized variance (RV) for day (t):
[
$$\text{RV}_t = \sum_{i=1}^{N_t} r_{t,i}^2$$

]
where $(r_{t,i})$ are intraday returns on intervals (e.g., 5-min). ([Economics Department](#))
- Use log-RV or winsorize RV for heavy tails.
- Practical note: for illiquid assets use realized kernels or subsampling.

Code / datasets: replicate RV computation from TAQ or public intraday sources (see *Data* below). Implement a small notebook to compute RV for SPY or a NASDAQ-100 symbol.

Why first: it grounds you in the measurement that every forecasting model targets; many mistakes come from noisy proxies.

2) Corsi — A Simple Approximate Long-Memory Model of Realized Volatility (HAR model) — Fulvio Corsi (2009) ([statmath.wu.ac.at](#))

What to learn:

- HAR model structure (intuitive heterogeneous time-horizon model), interpretation as additive contributions from traders with different horizons.

Key equation (HAR-RV):

[
$$\text{RV}_{t+1} = \beta_0 + \beta_1 \text{RV}_t^{(d)} + \beta_2 \text{RV}_t^{(w)} + \beta_3 \text{RV}_t^{(m)} + \varepsilon_{t+1}$$

]
where $(\text{RV}_t^{(w)} = \frac{1}{5} \sum_{i=0}^4 \text{RV}_{t-i}^{(d)})$, and similarly for month (21 days). ([statmath.wu.ac.at](#))

Replication tasks while reading:

- Implement HAR via OLS on your RV series (use log(RV) and raw RV variants).
- Try alternative lags (day/week/month vs day, week, month) and persistence diagnostics (ACF for log-RV).

Why second: HAR is the simple, industry-standard baseline — everything gets compared against HAR.

3) Hansen, Huang & Shek — Realized GARCH (joint returns + realized measure modelling) ([public.econ.duke.edu](#))

What to learn:

- Realized-GARCH measurement equation: link realized measure (x_t) and latent volatility (h_t) via a measurement equation; joint estimation improves fit vs GARCH alone.

Key model skeleton (log-linear RealGARCH):

- Return equation: $(r_t = \sigma_t z_t), (\sigma_t^2 = h_t)$.
- GARCH-type variance equation for (h_t) (ARMA style), and measurement equation:
[
$$x_t = \xi + \delta h_t + \tau z_t + u_t$$

]
(paper shows linear/log variants). (public.econ.duke.edu)

Replication tasks:

- Estimate a simple RealGARCH on index returns + RV (use R realizedGARCH packages or Python implementations). Compare out-of-sample to HAR and GARCH(1,1).

Why third: it's the bridge between classic volatility econometrics and realized-measure informed models.

4) Forecast evaluation basics — Patton (robust losses / QLIKE) and Diebold-Mariano tests

What to learn:

- **Loss choices:** MSE vs **QLIKE**. Patton shows MSE and QLIKE are robust choices; QLIKE is often preferred with noisy realized proxies. QLIKE (as used in Patton) for proxy $(\hat{\sigma}_t^2)$ and forecast (h_t) :
[
$$\text{QLIKE}(\hat{\sigma}_t^2, h_t) = \log h_t + \frac{\hat{\sigma}_t^2}{h_t}$$

]
(Patton shows QLIKE is homogeneous and robust to measurement noise).
- **Statistical tests:** Diebold-Mariano (DM) for forecast comparisons; MCS / SPA for multiple model comparisons. (jstor.org)

Replication tasks:

- Implement QLIKE and MSE evaluation functions; code Bootstrap/DM tests; implement the Model Confidence Set (MCS) or SPA (libraries exist).

Why fourth: proper evaluation avoids picking models that only 'look' better due to proxy noise.

5) HARNet — *HARNet: A Convolutional Neural Network for Realized Volatility Forecasting* (Reisenhofer et al., 2022) — paper + code ([arXiv](https://arxiv.org/abs/2205.12345))

What to learn:

- HARNet is a CNN/dilated-conv architecture that is explicitly initialized to reproduce the HAR baseline before training — this stabilizes optimization, especially with **QLIKE** loss. The model mimics HAR's multi-horizon receptive fields via dilated convolutions.

Architecture / formulas:

- Dilated causal convolution layers with increasing dilation factors ($d=1,2,4,\dots$) create exponentially growing receptive fields.
- Important practical trick: **HAR initialization** — set convolution weights so that pre-training predictions equal HAR OLS forecasts; then train to improve. ([arXiv](#))

Code: HARNet TensorFlow repo (authors' implementation) — use it as canonical starter. ([GitHub](#))

Replication tasks:

- Run HARNet on same dataset used in paper (authors provide configs and example dataset instructions).
- Try HARNet with (a) MSE (b) QLIKE — note optimization stability differences; replicate paper result that QLIKE + HAR init stabilizes training.

Why here: HARNet is a modern *HAR-aware* ML model — perfect hybrid baseline to compare against HAR.

6) DeepVol — *DeepVol: Dilated Causal Convolutions on High-Frequency Data* (Moreno-Pino & Zohren, 2022) — paper + code ([arXiv](#))

What to learn:

- Uses **intraday high-frequency returns directly** (not precomputed RV) as input to dilated causal convs (WaveNet-like) to predict next-day volatility. This can capture intraday patterns lost by pre-aggregation.

Architecture / formulas:

- Stacked dilated causal convolutions; residual blocks; dilations chosen to span intraday sequences. Hyperparameters: receptive field must cover relevant intraday lags. ([arXiv](#))

Code: VBayesLab/Deepvol (Jupyter notebooks implementing DeepVol) — good starting point. ([GitHub](#))

Replication tasks:

- Run the DeepVol training on a liquid stock index (NASDAQ-100 / AAPL) using 1- or 5-minute data. Compare: HAR (RV inputs) vs DeepVol (raw intraday inputs) vs HARNet (precomputed RV inputs). Evaluate with QLIKE and hedging P&L.

Practical note: HF data is large; subsample to 1- or 5-minute returns; be careful with market open/close microstructure noise.

7) Volt — *Volatility-based kernels & GP hierarchical models* (Benton, Maddox, Wilson; ICML 2022) — paper + code ([Proceedings of Machine Learning Research](#))

What to learn:

- Volt recasts stochastic volatility models as hierarchical Gaussian Process models with specialized kernels and mean functions (Magpie). It produces *calibrated predictive distributions* rather than point forecasts.

Key model pieces:

- Latent volatility path inferred and used as input to a GP for predictive distribution in observation space. See the paper for the SDE→GP derivation and the bespoke kernels. ([Proceedings of Machine Learning Research](#))

Code: g-benton/Volt repo (public implementation). ([GitHub](#))

Replication tasks:

- Try Volt for daily volatility forecasting on index data and compare *uncertainty calibration* (coverage of predictive intervals) vs DeepVol/HARNet point forecasts. Evaluate whether better calibration helps risk metrics (VaR hedging).

Why Volt: It gives uncertainty quantification and a principled stochastic-volatility inductive bias — great for original contributions pairing with ML models.

8) GNNHAR / Graph approaches — *Graph Neural Networks for multivariate realized vol* (Zhang et al. and code) ([arXiv](#))

What to learn:

- How to include **cross-asset spillovers** and relational structure (e.g., sector graphs, correlation networks) into HAR-style forecasting using GNNs and graph-based adjacency in multivariate HAR.

Key architecture idea / equation:

- GNN propagation step (example): $H^{(l+1)} = \text{ReLU}(\tilde{W} H^{(l)} \Theta^{(l)})$ with normalized adjacency $(\tilde{W})$. Use vectorized HAR inputs per node and allow multi-hop spillover. ([arXiv](#))

Code: chaozhang-ox / GNNHAR repo (reproducible code). ([GitHub](#))

Replication tasks:

- Run multivariate forecasts over, say, 50 stocks (SP500 subset). Compare univariate HAR/HARNet/DeepVol vs GNNHAR for short horizons. Test whether spillovers improve accuracy and economic utility (portfolio VaR, minimum-variance portfolio).

Why here: Many funds operate multivariate portfolios — showing spillover modeling is highly relevant.

9) Transformer / Multi-Transformer work (Ramos-Pérez et al., 2021) + other transformer variants ([arXiv](#))

What to learn:

- How attention/transformers can capture cross-scale patterns in volatility; multidimensional inputs (implied vol, macro, HF features). Use Multi-Transformer repo as starting code.

Replication tasks:

- Implement a transformer variant to forecast 1-day and 5-day RV using daily features (lagged RVs, VIX, volumes). Compare to HARNet/DeepVol.

Concrete step-by-step implementation guide (detailed, chronological)

Below: the **implementation sprints** (no time estimates — just ordered steps). For each step I list *commands / notebooks* to create, datasets, expected outputs, and checks.

A — Data & environment prep (notebook: 01_data_prep.ipynb)

1. **Pick assets:** start with 2× universes: (A) single liquid index (e.g., NASDAQ-100 / SPY / S&P500 index), (B) cross-section of 50–100 liquid stocks.
2. **Data sources (public / free):**
 - **Intraday tick/1-min/5-min:** Use L2/dealer data if available; otherwise use public minute data from **Kaggle** or **IEX Cloud** (free tier), **Kaggle: NYSE 1-min**, or the VBLab sample datasets. (Many academic repos ship processed intraday CSVs.) See DeepVol repo for dataset pointers. ([GitHub](#))
 - **Daily OHLC & returns:** Yahoo / yfinance for quick checks.
 - **Realized measure libraries:** VBayesLab datasets and (historically) Oxford-Man Realized Library (if available) — note the library was deprecated; use VBayesLab examples. ([Oxford Man Institute](#))
3. **Compute realized variance:** implement 5-min RV per Andersen formula; produce `rv_daily.csv`. (Notebook must include: sampling code, removal of overnight gaps, winsorization, log transform checks). ([Economics Department](#))
4. **Sanity checks:** verify RV autocorrelations, histogram, sample mean/variance.

Deliverable: directory `data/` with `rv_daily.csv`, `intraday_5min.parquet`, `returns_daily.csv`.

B — Baseline models (notebook: 02_baselines.ipynb)

1. **HAR (OLS) — univariate**
 - Fit HAR (day/week/month) on $\log(RV)$ and RV (both variants). Save parameters & residuals. Reproduce ACF & persistence.
 - Implement variant: HAR-X (include VIX or implied variance).
2. **GARCH(1,1) and Realized-GARCH**
 - Use `arch` (Python) or `rugarch` (R) to fit GARCH(1,1).
 - Fit Realized-GARCH (use Hansen et al. implementation or translate the model). ([public.econ.duke.edu](#))

Deliverable: outputs `models/har.pkl`, `models/garch.pkl`, prediction CSVs for in-sample & rolling OOS.

C — Evaluation & utilities (notebook: 03_evaluation.ipynb)

1. Implement **QLIKE** (Patton formula) and **MSE** and **MAE**. Use QLIKE for your main ranking.
 - QLIKE as implemented in Patton: $qlike = np.log(h) + rv / h$
2. Implement **Diebold-Mariano test** and the **Model Confidence Set** (MCS) wrapper. Use statsmodels DM test or custom bootstrap if required. ([jstor.org](https://www.jstor.org))
3. **Economic tests**: VaR backtests (Kupiec, Christoffersen) and hedging P&L: simulate delta-hedge P&L using predicted variance vs realized.

Deliverable: results/eval_table.csv and plotting utilities.

D — Implement HARNet / DeepVol / Volt / GNNHAR / Transformer (each in separate notebooks)

D1 — HARNet (notebook 04_harnet.ipynb)

- Clone/inspect authors' code (mdsunivie/HARNet). ([GitHub](https://github.com/mdsunivie/HARNet))
- Steps:
 1. Prepare HARNet inputs (same RV windows as HAR).
 2. Initialize HARNet with HAR weights (author code has HAR init).
 3. Train with QLIKE and MSE - compare training curves & stability.
 4. Output OOS forecasts and save model.

Pitfall & tip: QLIKE optimization can be unstable; use HAR initialization + smaller LR; monitor gradient norm. ([arXiv](https://arxiv.org))

D2 — DeepVol (notebook 05_deepvol.ipynb)

- Use VBayesLab/Deepvol notebooks as starting point. ([GitHub](https://github.com))
- Steps:
 1. Construct intraday sequences (e.g., 390 minutes for US equity days, or subsample to 5-min → 78 values).
 2. Train dilated causal convs (WaveNet style) to predict next-day RV or variance.
 3. Compare with HARNet and HAR using QLIKE and VaR hedging.

Tip: use smaller batch sizes for long sequences; consider channel normalization per day.

D3 — Volt (notebook 06_volt.ipynb)

- Clone Volt repo and run examples. ([GitHub](https://github.com))
- Steps:
 1. Use daily RV series as input.
 2. Fit Volt and extract predictive intervals; calibrate coverage (e.g., 95% intervals).

3. Compare coverage and distributional metrics vs deep models.

Contribution angle: combine Volt predictive intervals with HARNet/DeepVol point forecasts into a *calibrated ensemble*.

D4 — GNNHAR (notebook 07_gnnhar.ipynb)

- Clone chaozhang-ox/GNNHAR and run reproducible scripts. ([GitHub](#))
- Steps:
 1. Build adjacency (correlation threshold, sector graph, or graphical LASSO).
 2. Train GNNHAR on multivariate RV data.
 3. Compare cross-sectional forecasts & show whether spillovers help.

D5 — Transformer / Multi-Transformer (notebook 08_transformer.ipynb)

- Use MultiTransformer repo for code patterns. ([GitHub](#))
- Steps:
 1. Train transformer on daily features; test multi-horizon forecasts.
 2. Compare to HARNet and DeepVol.

E — Experiments & ablations (notebook: 09_experiments.ipynb)

Design a controlled experiment grid:

Model variants (rows): HAR, HAR + X (VIX), RealGARCH, HARNet, DeepVol, Volt (GP), GNNHAR, Transformer.

Inputs (columns): daily RV only / RV+ImpliedVol / HF raw intraday / multivariate cross-section.

Loss and eval: QLIKE (primary), MSE (secondary), VaR hedging P&L, interval calibration metrics (for Volt), DM test / MCS for significance.

Ablations (must include):

- HF vs precomputed RV (DeepVol vs HARNet).
- HAR initialization vs random init for HARNet. ([arXiv](#))
- QLIKE vs MSE training loss for deep models.
- Ensemble weighting (stacking) vs single model.

Statistical testing:

- For each pair (model A vs B) compute DM test on QLIKE loss differential. Use bootstrap if loss differential shows high autocorrelation. ([jstor.org](#))

Economic tests:

- Hedge simulated option delta using predicted variance; compare realized P&L, Sharpe improvements, and VaR backtests.

Deliverables: big table results/aggregate.csv with QLIKE, MSE, VaR results, DM p-values, coverage stats.

F — Robustness / regime / sanity checks (notebook: 10_robustness.ipynb)

1. **Regime splits:** bull vs bear periods (e.g., 2020 COVID spike vs calm 2017); report model ranking per regime.
 2. **Liquidity tests:** compare on small-cap vs large-cap stocks.
 3. **Sensitivity:** to sampling frequency (1-min vs 5-min) and RV estimator (RV vs realised kernel).
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G — Writeup, reproducibility & GitHub (notebook: 11_writeup.ipynb + README.md)

- Produce a paper-style README + paper.pdf with (i) Abstract, (ii) Methods, (iii) Experiments, (iv) Results & Figures, (v) Conclusion & suggested extensions. Include reproducibility manifest (seed, env YAML, data retrieval steps). Use the paper templates from HARNet/DeepVol teams to match academic style. ([GitHub](#))

Repo structure suggestion

```
vol_forecast_project/  
  data/  
  notebooks/ # 01_data_prep.ipynb ... 11_writeup.ipynb  
  src/  
    models/  
    utils/  
  results/  
    figures/  
    tables/  
  environments/  
    conda_env.yml  
  README.md  
  LICENSE
```

Paper-by-paper “important equations / concepts” cheat-sheet (extract these while reading each paper)

- **Andersen et al. (2001):** realized variance definition and microstructure caveats. ([Economics Department](#))

- **HAR (Corsi 2009)**: HAR additive model; compute weekly/monthly aggregations; log-transform options. (statmath.wu.ac.at)
- **Realized-GARCH (Hansen et al.)**: measurement equation linking realized measure and latent variance; joint estimation. (public.econ.duke.edu)
- **Patton (Robust losses)**: QLIKE formula: $QLIKE = \log(h) + RV / h$ (use RV as proxy) and theoretical justification for using QLIKE with noisy proxies.
- **Diebold-Mariano (1995)**: test statistic for equal predictive ability (use with QLIKE differentials; beware autocorrelation). (jstor.org)
- **HARNet (Reisenhofer et al.)**: dilated convs, HAR initialization trick, QLIKE training stability. ([arXiv](https://arxiv.org))
- **DeepVol (Moreno-Pino & Zohren)**: raw HF→dilated causal convs (WaveNet style) to produce day-ahead volatility — architecture hyperparams (dilations, receptive field). ([arXiv](https://arxiv.org))
- **Volt (Benton et al.)**: hierarchically derived GP kernel from stochastic volatility SDEs; Magpie mean functions for trend. Focus on predictive distribution & calibration. ([Proceedings of Machine Learning Research](https://proceedings.mlr.press/))
- **GNNHAR (Zhang et al.)**: adjacency construction, multi-node HAR extension, graph convolutions for spillovers. ([arXiv](https://arxiv.org))
- **Multi-Transformer (Ramos-Pérez)**: transformer attention adapted for volatility forecasting; code available. ([arXiv](https://arxiv.org))

Concrete final experiment checklist (so you can mark DONE / TODO)

- Data: compute RV (5-min, 1-min) for index + 50 stocks. (notebook 01_data_prep.ipynb) (Economics Department)
- Baselines: HAR (OLS), GARCH(1,1), Realized-GARCH. (02_baselines.ipynb) (statmath.wu.ac.at)
- Evaluation: implement QLIKE (primary), DM tests, MCS. (03_evaluation.ipynb)
- ML Models: HARNet (authors' repo), DeepVol (VBLab repo), Volt (g-benton/Volt), Transformer (Multi-Transformer), GNNHAR (chaozhang-ox). (04-08_*.ipynb) ([GitHub](https://github.com))
- Ablations: HF vs RV, QLIKE vs MSE, HAR init vs random, multivariate spillover. (09_experiments.ipynb)
- Robustness: regime splits, liquidity splits. (10_robustness.ipynb)
- Writeup & figures for GitHub + draft paper. (11_writeup.ipynb)

Good original contributions (pick 1–2 to pursue in the repo / paper)

1. **Volt + DeepVol hybrid**: use Volt to provide calibrated predictive intervals for DeepVol point forecasts (learn a mapping from DeepVol features to Volt latent volatility prior) — show improved VaR hedging and coverage. (Novelty: ML point predictions + GP calibrated uncertainty.) ([Proceedings of Machine Learning Research](https://proceedings.mlr.press/))

2. **HARNet + GNN cross-section:** extend HARNet inputs with GNN-derived cross-sectional spillover features (learned adjacency), show benefits for portfolio VaR. ([arXiv](#))
3. **Pretrained universal volatility model (DeepVol style) + few-shot fine-tuning** across assets — demonstrate transfer learning for low-liquidity assets (train on large liquid stocks, fine-tune on small caps). (VBLab has related work you can cite.) ([GitHub](#))

Practical tips / gotchas (from the literature + implementation experience)

- **Always log or winsorize RV** before regression/ML — heavy tails can break training. ([Economics Department](#))
- **QLIKE** is preferable for ranking with noisy RV proxies; *train deep models with QLIKE when your goal is better economic risk measures*. Patton shows QLIKE is robust.
- **HAR initialization** strongly stabilizes training with QLIKE (HARNet result). If you train HARNet from random init with QLIKE you may see poor convergence. ([arXiv](#))
- **HF data size:** use 5-min as practical compromise (reduced microstructure noise + reasonable sequence length). DeepVol papers used Nasdaq-100 intraday two-year windows — replicate with smaller pilot dataset first. ([arXiv](#))
- **Statistical tests:** DM has reduced power under strong serial dependence — use block-bootstrap corrections / MCS for multiple comparisons. ([NBER](#))

Quick list of the core papers + code (click the citations to open PDF / repo)

1. **HAR (Corsi)** — *A Simple Approximate Long-Memory Model of Realized Volatility* (Corsi, 2009). Paper (PDF). ([statmath.wu.ac.at](#))
2. **Andersen et al.** — *Modeling and Forecasting Realized Volatility* (Andersen, Bollerslev, Diebold & Labys). Paper (PDF). ([Economics Department](#))
3. **Realized-GARCH** — Hansen, Huang & Shek (2011/2012). Paper (SSRN / PDF). ([public.econ.duke.edu](#))
4. **QLIKE / evaluation** — Patton (robust loss / forecast eval). Paper (PDF).
5. **Diebold & Mariano** — *Comparing Predictive Accuracy* (DM Test). Paper. ([jstor.org](#))
6. **HARNet** — *HARNet: A Convolutional Neural Network for Realized Volatility Forecasting* (Reisenhofer et al., 2022). Paper + code repo. ([arXiv](#))
7. **DeepVol** — *DeepVol: Volatility Forecasting from High-Frequency Data* (Moreno-Pino & Zohren, 2022/24). Paper + VBayesLab implementation. ([arXiv](#))
8. **Volt (GP)** — *Volatility Based Kernels and Moving Average Means for Accurate Forecasting with Gaussian Processes* (Benton, Maddox & Wilson, ICML 2022). Paper + repo. ([Proceedings of Machine Learning Research](#))
9. **GNNHAR / Graph methods** — *Graph Neural Networks for Forecasting Multivariate Realized Volatility* (Zhang et al.) + code. ([arXiv](#))

10. **Multi-Transformer** — *Multi-Transformer for S&P Volatility (Ramos-Pérez et al.)* + code. ([arXiv](#))
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Final advice — what to deliver to impress quant recruiters

- A **clean GitHub repo** with: reproducible notebooks, conda/pip env file, scripts to reproduce figures, and a paper.pdf summarizing methods + results. Use the HARNet/DeepVol repos as a style guide for reproducibility. ([GitHub](#))
 - **Key figures to include:** QLIKE ranking table, DM p-values matrix, coverage plots (Vola), hedging P&L cumulative curves, regime-split performance table, and filter/feature-importance visualizations for ML models.
 - **Writeup:** include clear economic interpretation — e.g., “Model X reduces one-day VaR breaches by Y% vs HAR” — recruiters care about practical risk improvement, not only lower MSE.
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If you want, I can now:

- (A) produce a **skeleton GitHub repository** (file tree + README + environment YAML + one working notebook to compute RV) that implements the first two sprints, or
- (B) generate the exact **experiment matrix CSV** and a checklist of notebooks + function prototypes you can copy/paste.

Which of (A) or (B) would you like me to produce next? (I can create files / notebooks structure for you immediately.)